MANMEET SINGH

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PROFESSIONAL SUMMARY

Full Stack Developer with a strong foundation in web development and Data Structures & Algorithms. Skilled in building responsive and scalable applications using modern technologies. Passionate about problem-solving, developing efficient solutions, and continuously learning new tools and frameworks to create impactful software.

TECHNICAL SKILLS

Languages: C++, JavaScript, Python, SQL

Backend: Node.js, Express.js, RESTful APIs

Frontend: React.js, Tailwind CSS, Bootstrap, HTML5, CSS3, Figma

Database: MySQL, MongoDB, Firebase

Automation: n8n

AI Tools And API: NVIDIA NIM

Cloud & DevOps: Git, GitHub, Docker, Vercel, Render, Brevo SMTP, Resend

Data Structure and Algorithm (Core Concepts): C++

PROFESSIONAL EXPERIENCE

Data Science Intern | DUCAT

May 2025 - July 2025

Ludhiana, India

- Built and evaluated an XGBoost regression model to predict dish ratings from ingredient data using a dataset of 1000 recipes
- Designed a data preprocessing pipeline with ingredient cleaning and TF-IDF vectorization for feature extraction
- Achieved **RMSE of 0.63 and MAE of 0.52** on a 5-point rating scale, indicating strong prediction accuracy
- Developed an evaluation script to replicate preprocessing and **automate model validation using Scikit-learn metrics**
- Set up the ML environment and dependencies (**XGBoost, Scikit-learn**) to ensure reproducibility and model consistency

PROJECTS

Roofing Services Website |

Jan 2026 – Feb 2026

React.js, Node.js, Express.js, MongoDB, Tailwind CSS

- *Developed a responsive full-stack website for a roofing services business to showcase services and generate customer leads*
- *Built dynamic frontend using React.js and Tailwind CSS for a modern and user-friendly interface*
- *Implemented backend using Node.js and Express.js to handle contact forms and customer inquiries*
- *Integrated database (MongoDB) to store and manage user queries efficiently*
- *Focused on mobile responsiveness and performance optimization for better user experience*

FarmSense AI – Smart IoT Irrigation System |

Mar 2026 – Apr 2026

ESP32, Firebase, React Native, IoT Sensors, AI

- *Developed an IoT-based system to monitor soil moisture, temperature, and control irrigation in real time*
- *Built ESP32-based hardware to collect sensor data and sync with Firebase cloud database*
- *Implemented automated irrigation using moisture thresholds and smart decision logic*
- *Designed a React Native app with real-time dashboard and multilingual AI assistant*
- *Enabled offline manual control via OLED display and joystick for pump operation*
- *Added data visualization and marketplace feature for sustainable crop residue management*

CERTIFICATIONS & ACHIEVEMENTS

- **GNA Hackathon 4.0 [Finalist]** | Built Personalized Learning Platform for student with cross platform for Teachers and Parents to track performance.
- **Udemy Certified Web Developer** - Udemy
- **Winner - Athena Hackathon 2024 (College Level)** | Built cheating free website for giving online test
- **Top 15 Teams | Most Innovative Idea (Agri-Tech) – CT Hackathon (National Level)** | FarmSense AI – Smart IoT Irrigation System

EDUCATION

B.Tech in Computer Science

2023 - 2027

Punjab College of Technical Education, Ludhiana, Punjab | CGPA: 7.4/10.0